9C Lesson 4

Treating diseases

Year 9 Science · 9C Threat from disease · Student Book pages 44 to 45

Name _____ Class _____ Date _____

Today I can:

☐ I can explain which medicines treat which types of disease, and why antibiotics do not work on viruses.

☐ I can explain how large amounts of human hormones are made quickly using genetically modified bacteria in a fermenter.

Key words

drug a substance that affects how the body works
medicine a drug used to treat or prevent disease
antibiotic a medicine that kills bacteria
antiviral a medicine that stops viruses replicating (copying themselves)
genetic disorder a disease caused by faulty genes that a person is born with
autoimmune disease a disease in which white blood cells attack the body's own cells
insulin a hormone made in the pancreas that lowers blood glucose
transgenic contains DNA from more than one species
fermenter a large container in which microorganisms are grown
sterilise kill all the microorganisms in or on something
aseptic free from unwanted microorganisms

Core Everyone does this section.

1. Match each scientific word to its definition. Draw a line.

medicines	drugs that prevent viruses replicating
antibiotics	a hormone that makes cells take up glucose from the blood
antivirals	drugs used to treat or prevent disease
vaccines	substances that make us immune to particular diseases
insulin	drugs that kill bacteria

2. Tick one box in each row to show the best treatment for each disease.

Disease	antibiotic	antiviral	add a nutrient	inject a hormone	vaccine (prevents it)
ear infection caused by bacteria					
flu (caused by a virus)					
scurvy (too little vitamin C)					
type 1 diabetes (no insulin made)					
measles, before a child catches it					

3. Complete the text about fermenters using words from the word bank.

Transgenic bacteria are grown in a large container called a _____. Before use it is _____ with steam so that no other _____ can grow inside it. _____ air is bubbled in so that the bacteria can respire. _____ paddles mix the bacteria, nutrients and

oxygen, and a water _____ keeps the temperature steady.

●● Stretch When Core is finished.

S1. Explain why medicines are drugs.

S2. Describe how transgenic bacteria are made to produce human growth hormone.

S3. In type 1 diabetes the body stops making insulin. Explain why this can cause organ damage, and state how the disease is treated.

●●● Challenge Explain in full sentences.

C1. Transgenic bacteria are grown in fermenters.

a) Give the reason why air is bubbled into the fermenter.

b) Describe two ways in which the fermenter keeps the bacteria growing at a high rate.

c) Describe the aseptic precautions taken when preparing the fermenter, and explain why they are needed.

C2. Scientists can also remove genes from an organism. Explain how this can create a genetically modified organism that is not transgenic.

My thinking (any language)

Some bacteria are no longer killed by antibiotics that used to work. Suggest why using antibiotics when they are not needed could make this problem worse.

Word bank drug · medicine · antibiotic · antiviral · vaccine · insulin · transgenic · fermenter · sterilised · microorganisms · Filtered · Stirring · jacket

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Exit ticket 9C Lesson 4: Treating diseases Name _____

1. What is the difference between an antibiotic and an antiviral?

2. What does 'transgenic' mean?

3. Give one reason why a fermenter is sterilised before use.

Today I can: Goal 1 ☐ Goal 2 ☐